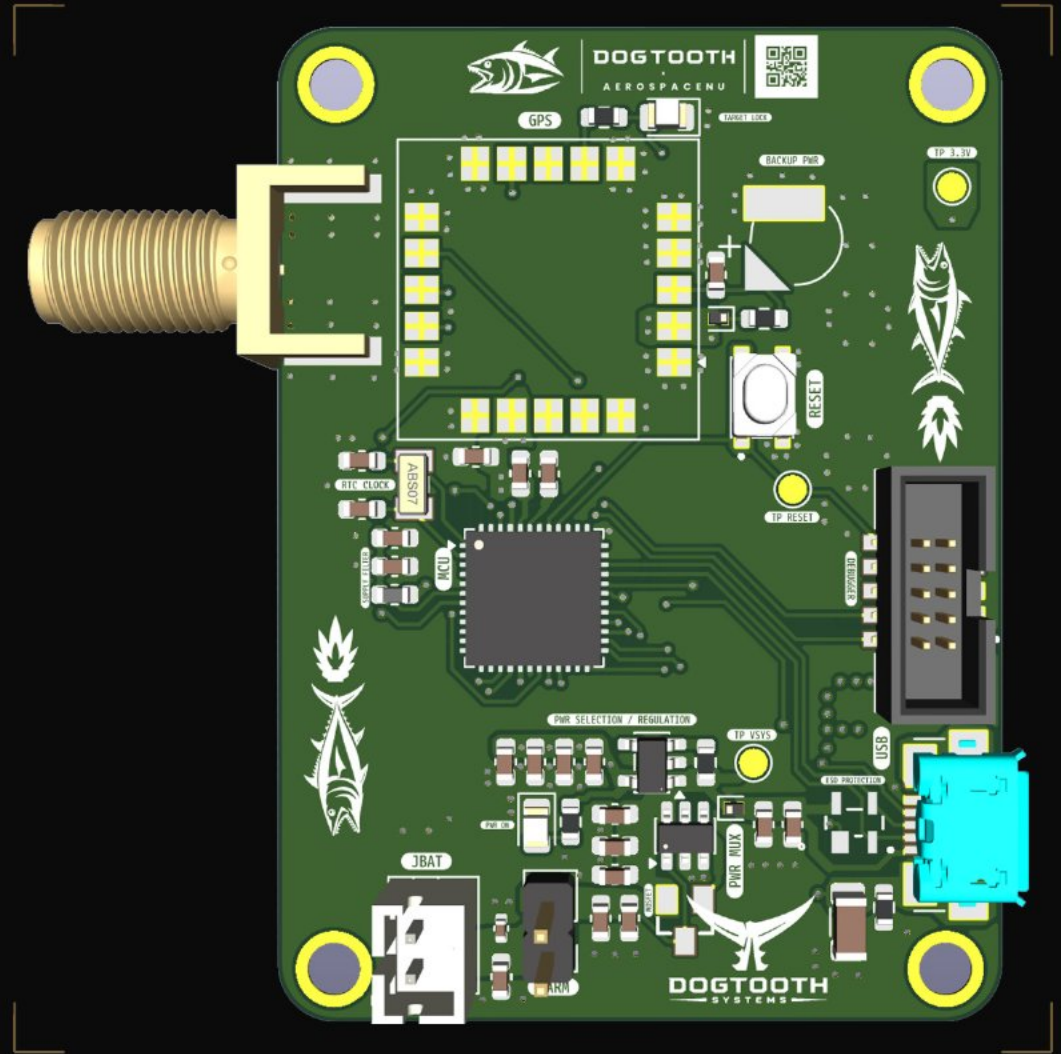


Custom Avionics GPS / Radio PCB

A custom KiCad printed circuit board for rocket GPS position acquisition and a 900 MHz telemetry downlink.

SAMD21 · u-blox SAM-M8Q · E22 900 MHz · 1S LiPo / USB



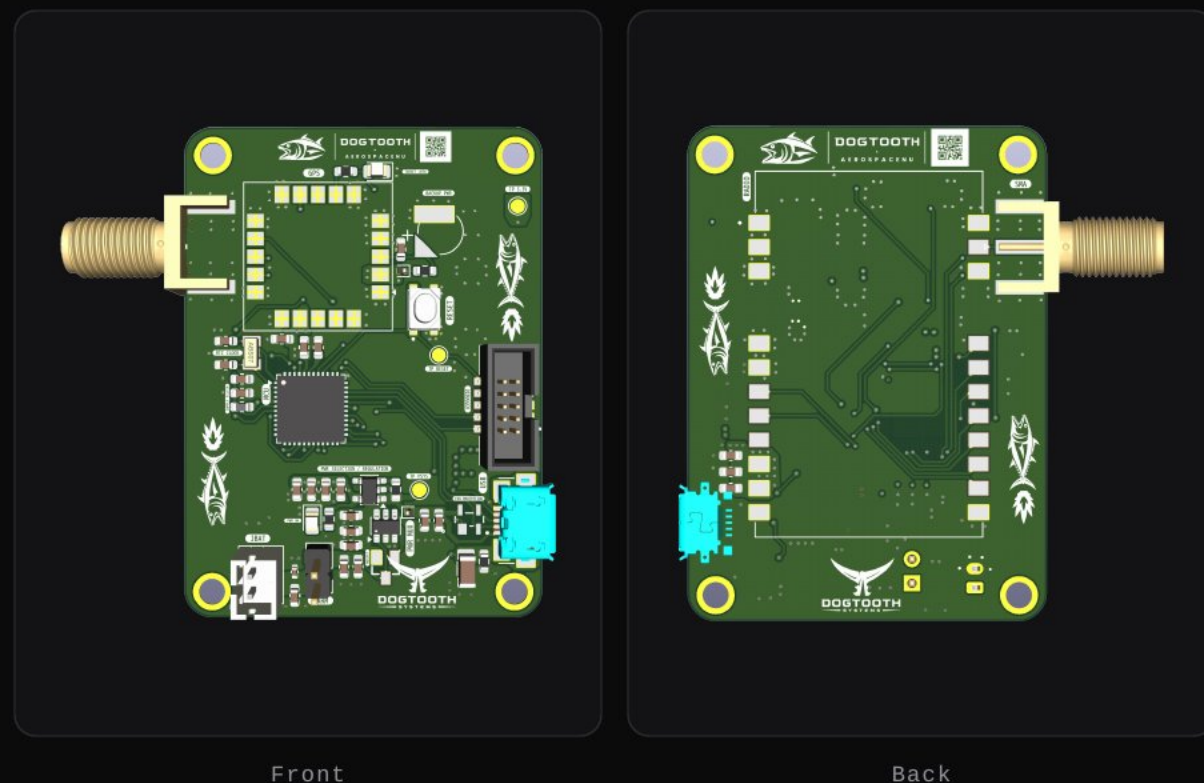


CONTENTS

What's inside

The finished board, front and back, and a guide to each section.

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01

PROJECT SNAPSHOT

System at a glance

Six building blocks define the board — the microcontroller, the GPS receiver, the radio, the power architecture, the external interfaces, and the toolchain.



MCU

ATSAMD21G18A — 48-pin Arm Cortex-M0+ microcontroller.



GPS

u-blox SAM-M8Q — integrated GNSS module with antenna.



RADIO

E22-900M30S1 — 900 MHz, SX1262-based LoRa module.



POWER

1S LiPo or USB bench power, regulated to +3.3 V.



INTERFACES

USB · UART / I²C · SPI radio · SWD debug.



TOOLS

KiCad 9.0.7 — hierarchical schematic & PCB layout.

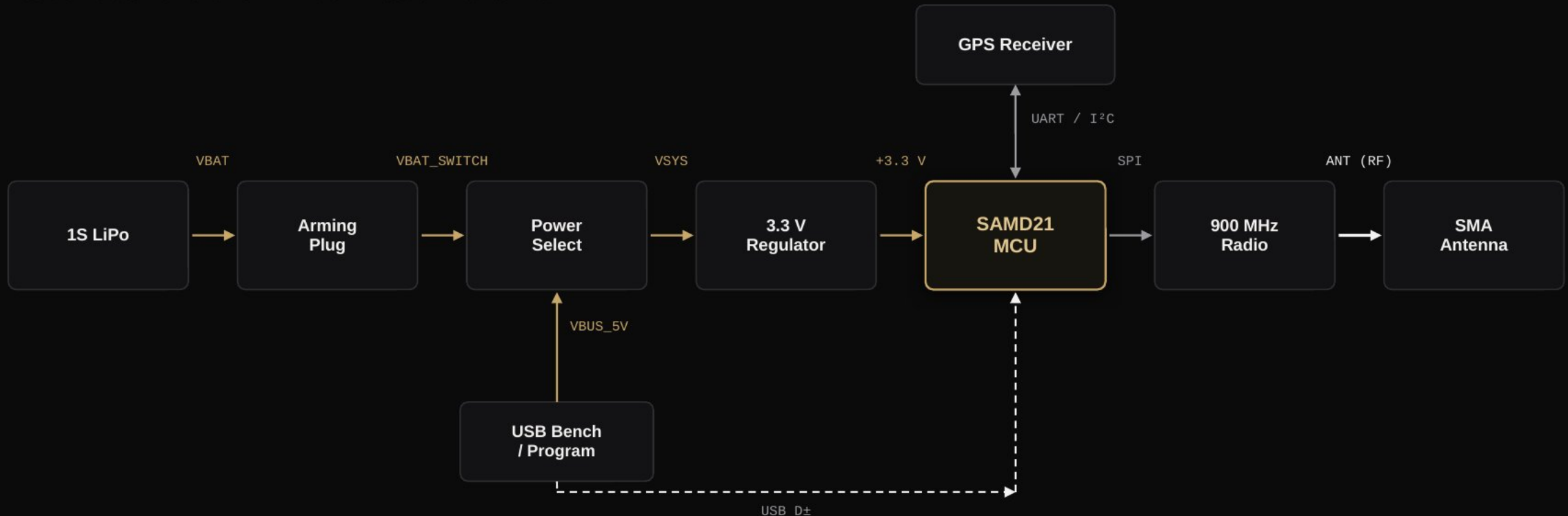


02

SYSTEM ARCHITECTURE

Power, data & RF paths

Everything is organised around the SAMD21. Power enters from the LiPo or USB, is selected and regulated to +3.3 V, then feeds the GPS and radio. Position data flows GPS → MCU → radio → antenna.

**POWER PATH**

VBAT · VBAT_SWITCH · VSYS · +3.3 V

DATA PATH

GPS ↔ MCU (UART/I²C) · MCU → radio (SPI) · USB D±

RF PATH

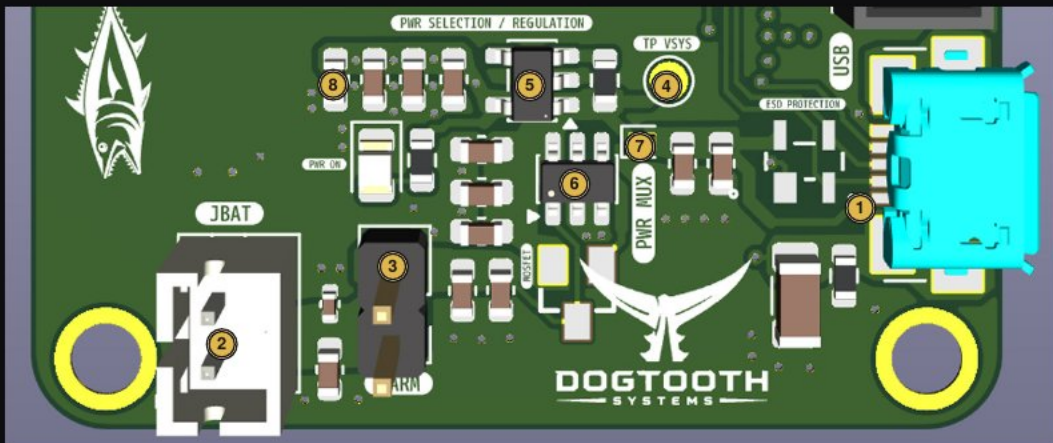
radio ANT → SMA · 50 Ω single-ended



03 POWER SYSTEM

Battery, USB, selection & regulation

Raw battery, battery-after-arm, USB 5 V, the selected system rail, and the regulated +3.3 V logic rail are kept distinct — so the board runs safely on USB at the bench while still supporting battery use in flight.



3D render: battery, arming plug, power switching, regulator and USB.

DESIGN NOTE

Bench-safe on USB, flight-ready on battery — the mux hands over automatically and the arming plug isolates the cell until launch.

1

VBUS_5V

Power from USB for bench testing.

2

VBAT

Battery power in, at connector J1.

3

VBAT_SWITCH

Battery power once armed (plug J2).

4

VSYS

Chosen power, sent to the regulator.

5

AP2112K-3.3

Holds power at a steady 3.3 V (U3).

6

LTC4412 + Q1

Picks USB or battery automatically.

7

D1 Schottky

Stops battery power leaking into USB.

8

+3.3 V rail

Powers the processor, GPS and radio.

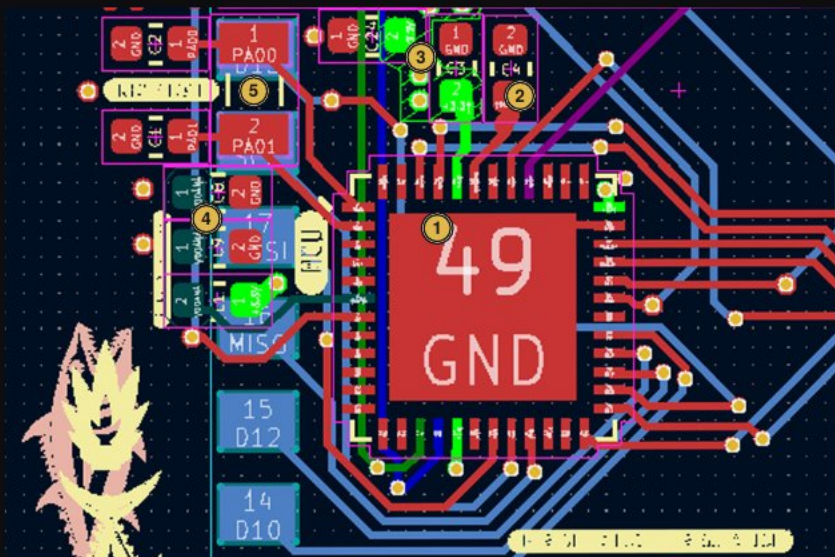


04

SAMD21 MCU SECTION

The control center

The ATSAM21G18A coordinates everything. It was chosen for ample GPIO, the common serial buses, native USB, and genuinely low-power embedded operation.



1

ATSAMD21G18A

Runs the code and controls every other part.

2

VDDCORE cap

Keeps the chip's internal power steady.

3

VDDIN / VDDIO

Capacitors that filter power noise.

4

VDDANA filter

Cleans power for the chip's analog circuits (L1).

5

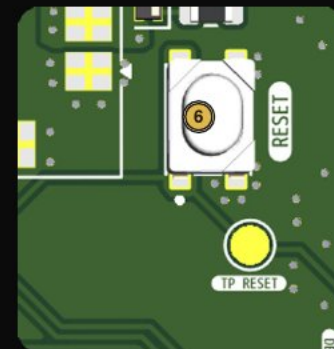
RTC crystal Y1

Keeps accurate time, like a watch crystal.

6

Reset support

Button (SW1) and test point to restart the chip.

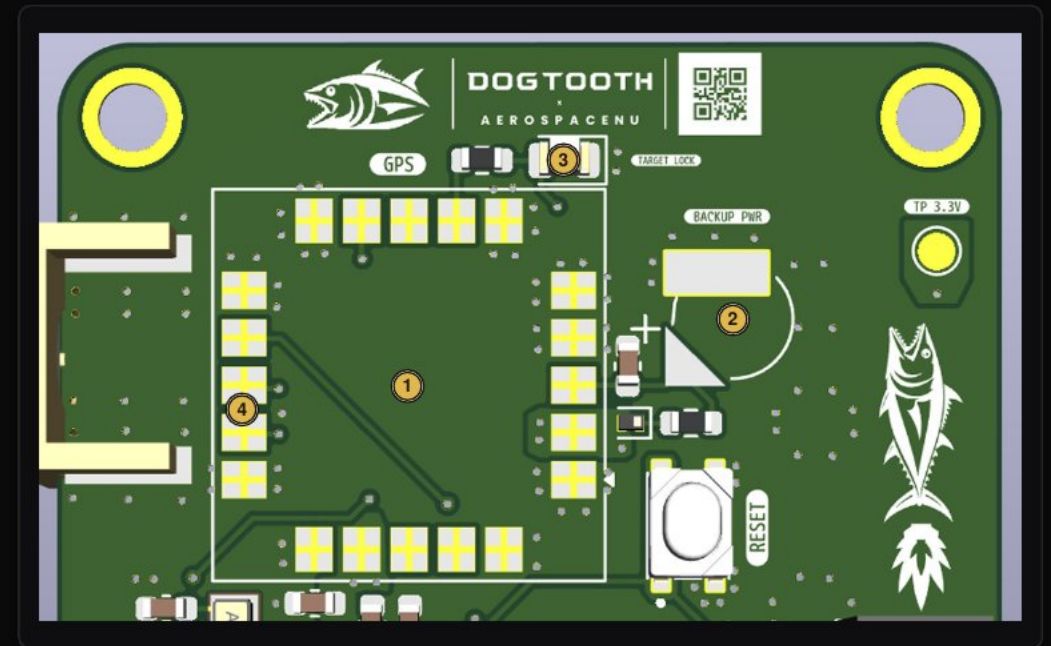




Position & time receiver

The u-blox SAM-M8Q acquires satellite position and timing. The layout keeps the receiver recognisable and groups the backup supply, status circuitry and data lines close to the module.

- 1 SAM-M8Q module**
Finds position and time using satellites.
- 2 V_BCKP backup**
Small backup cell that saves satellite data so GPS relocks faster.
- 3 TIMEPULSE + LED**
Timing signal, plus an LED (D4) that shows GPS lock.
- 4 UART / I²C**
Sends location data to the microcontroller.

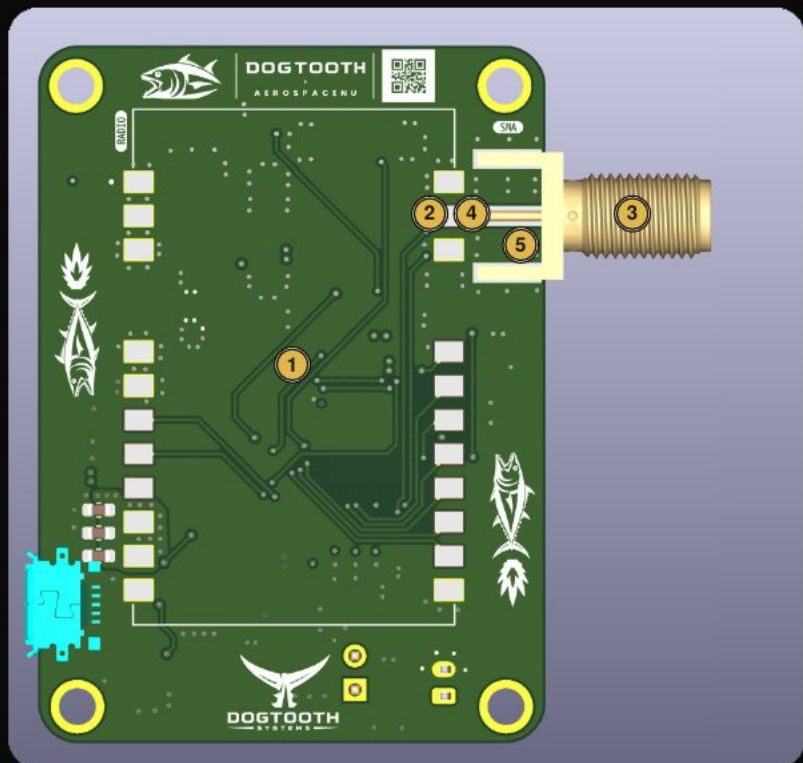


3D render: GPS module, backup cell, lock LED and reset button.



900 MHz link & antenna path

The E22-900M30S1 sends GPS-derived telemetry over a 900 MHz link. The module ANT pin routes to the SMA jack through a single-ended, impedance-controlled RF trace.



- 1 E22-900M30S1**
900 MHz radio that sends data to the ground.
- 2 ANT pin**
Where the radio signal leaves the module.
- 3 SMA connector**
Screw-on jack for the antenna.
- 4 50 Ω trace**
Sized trace that carries the signal cleanly.
- 5 Ground vias**
Ground links that keep the signal clean.

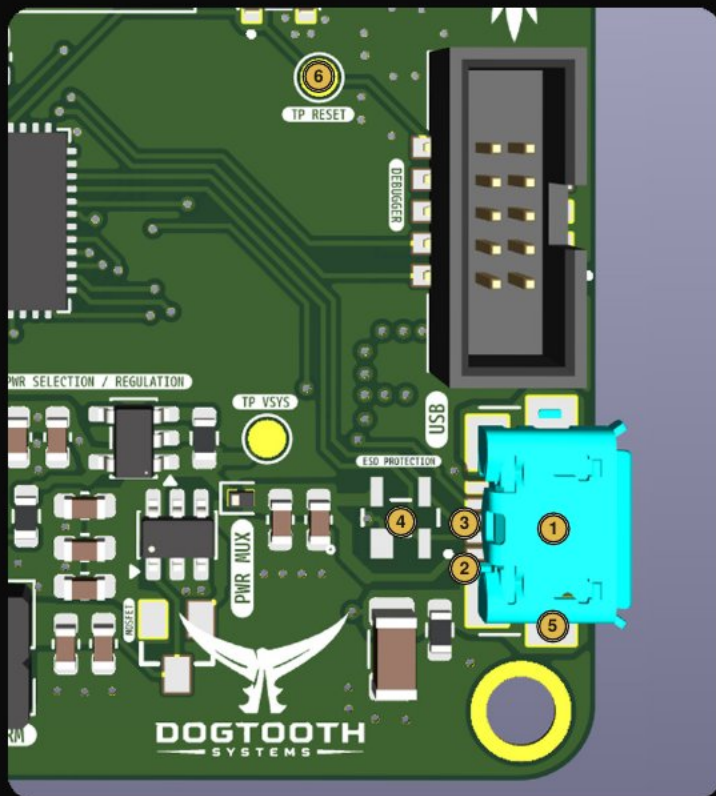




07 USB · ESD · SHIELD · DEBUG

Bench access & programming

The USB port provides bench power and a programming link. A dedicated ESD device protects D+/D-, and a separate SHIELD net keeps the connector shell off the logic ground.



1 USB connector J3

Micro-USB port for power and programming.

2 VBUS_5V

5 V power from the USB cable.

3 USB_D+ / D-

USB data lines to the microcontroller.

4 PRTR5V0U2X ESD

ESD protection for the USB data lines.

5 SHIELD net

Keeps the USB shell off circuit ground.

6 ~RESET line

Lets the debugger restart the chip.

SWD DEBUG HEADER

Amphenol 20021521 · Cortex-M 10-pin

+3.3 V reference / target power

GND signal ground

SWDIO serial-wire data

SWCLK serial-wire clock

~RESET target reset



08

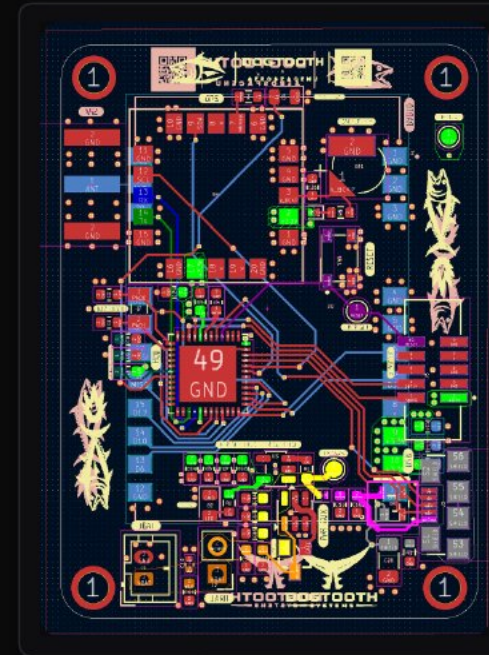
WHAT THIS DEMONSTRATES

Skills & engineering takeaways

What the project proves today — and the honest next steps before it could claim flight readiness.

SKILLS DEMONSTRATED

- ✓ Hierarchical schematic capture across power, MCU, GPS/radio and debug sheets in KiCad.
- ✓ Mixed-signal routing combining GPS, a 900 MHz radio, USB and battery power.
- ✓ Copper zones used for ground and power distribution.
- ✓ RF-aware layout with a 50 Ω single-ended antenna route.
- ✓ Decoupling, ground pours, stitching vias, silkscreen and DRC clean-up.





09

NEXT VALIDATION STEPS

Prototype fabrication plan

A staged path from design files to a brought-up board — each gate de-risks the next.





Custom Avionics GPS / Radio PCB

DESIGNED IN KICAD 9.x

This document does not claim flight-proven performance.

CHRISTOPHER SPARROW

**A**

APPENDIX

Source files & layers

The KiCad schematic, copper-layer views, stack-up and 3D renders behind the board.

SCHEMATIC

Four hierarchical sheets — root & MCU, power, GPS · radio · UART, and SWD debug.

COPPER LAYERS

Front and back copper with silkscreen, plus both layers combined.

INNER LAYER

The L3 GND pour kept under the SMA antenna route.

FOUR-LAYER STACK

Cross-section of the copper / dielectric stack-up and layer assignment.

3D RENDERS

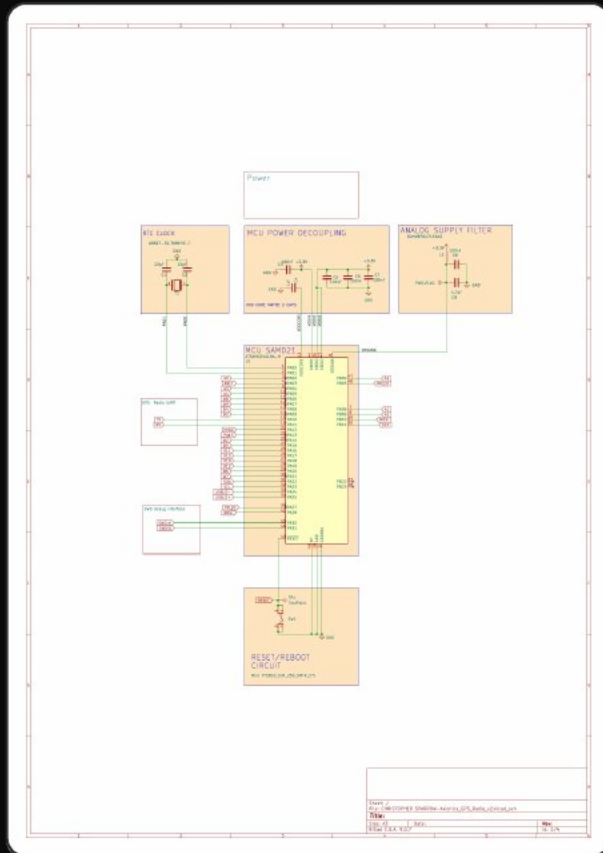
Assembled board, front and back, from the KiCad 3D viewer.



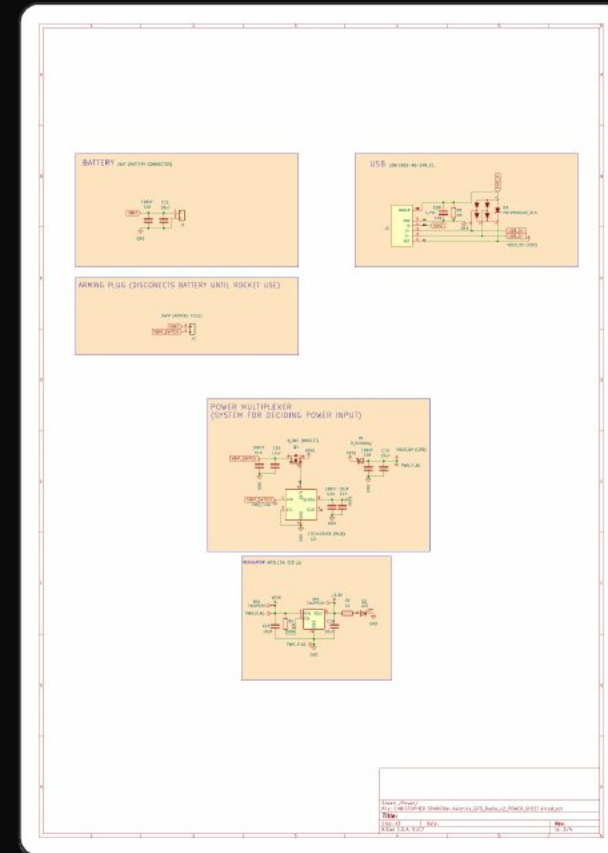
A SCHEMATIC

Schematic — root & power

SAMD21 core, decoupling, RTC and reset (left); battery, mux, regulator and USB (right).



Sheet 1 · Root & MCU



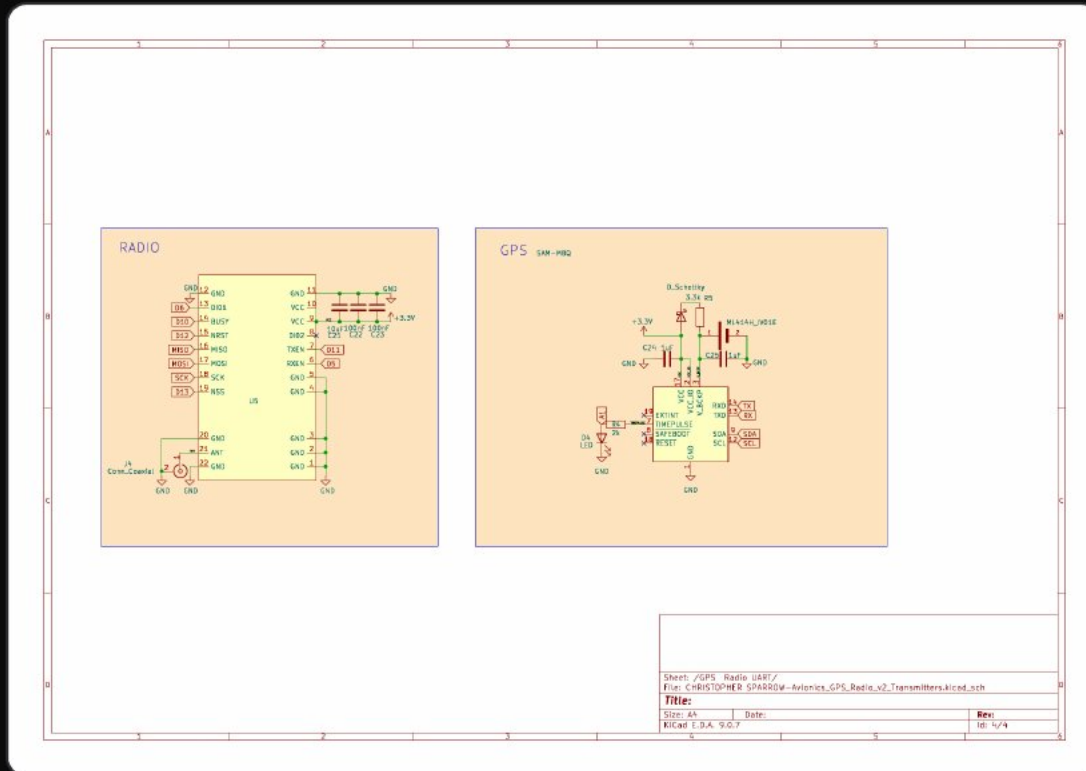
Sheet 2 · Power

**A**

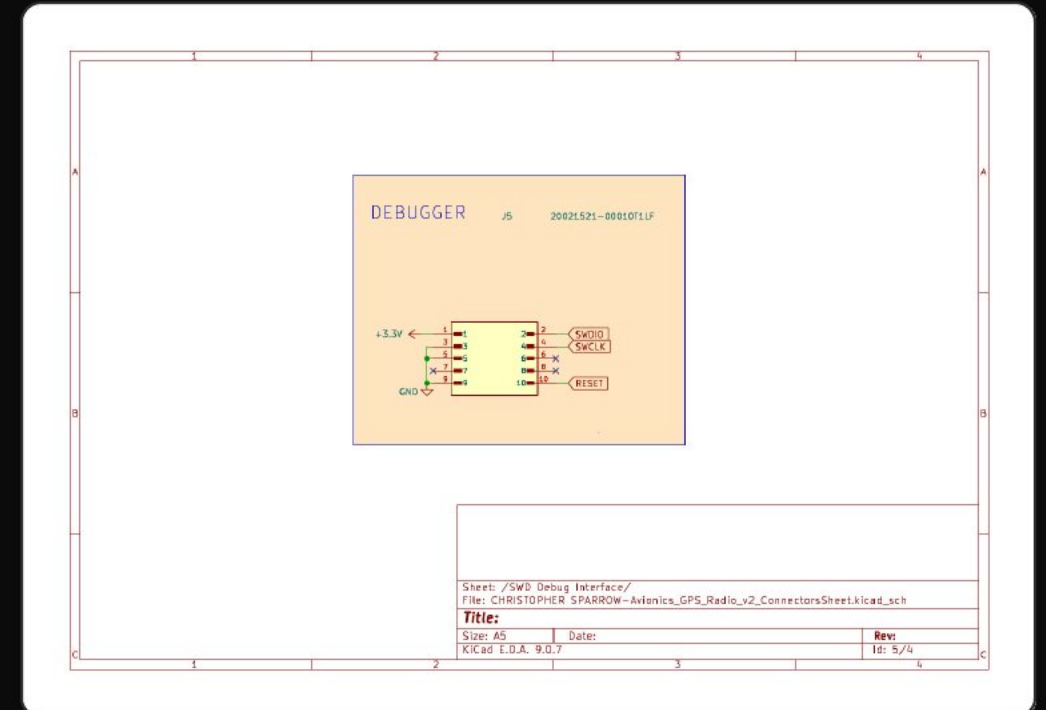
SCHEMATIC

Schematic — GPS, radio & debug

SAM-M8Q, E22 radio and backup cell (left); Cortex-M SWD header (right).



Sheet 3 · GPS · Radio · UART



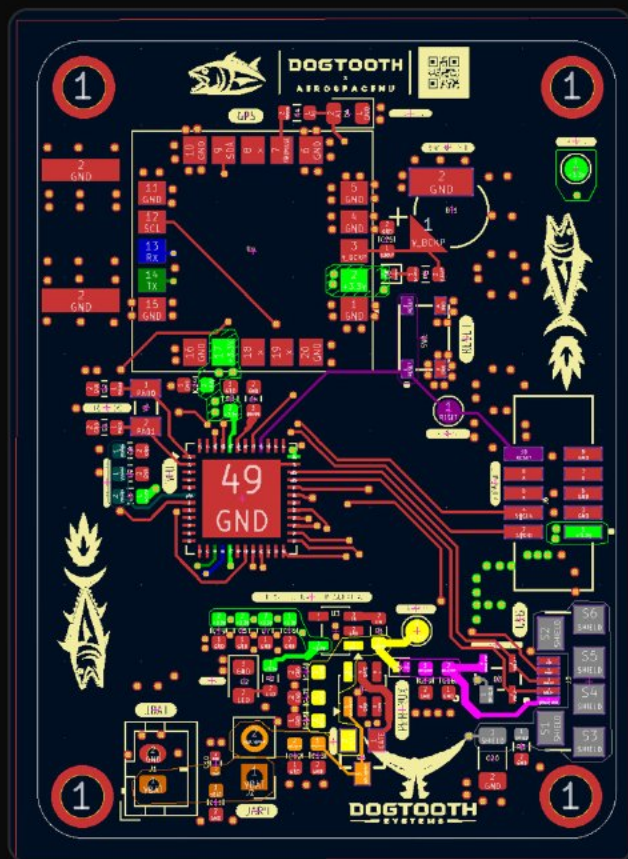
Sheet 4 · SWD debug



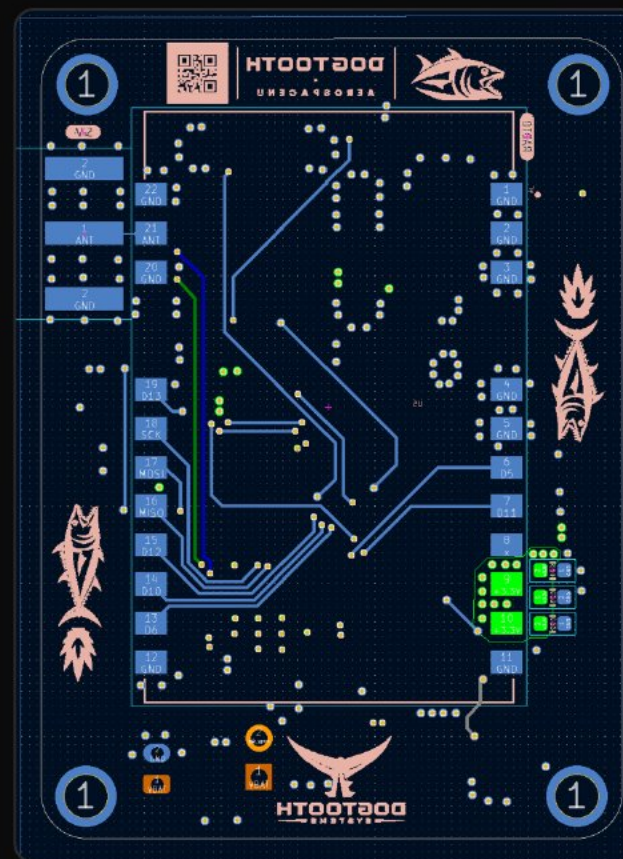
A COPPER LAYERS

Front & back copper

Each copper side with its silkscreen, zones unfilled.



Front · zones unfilled



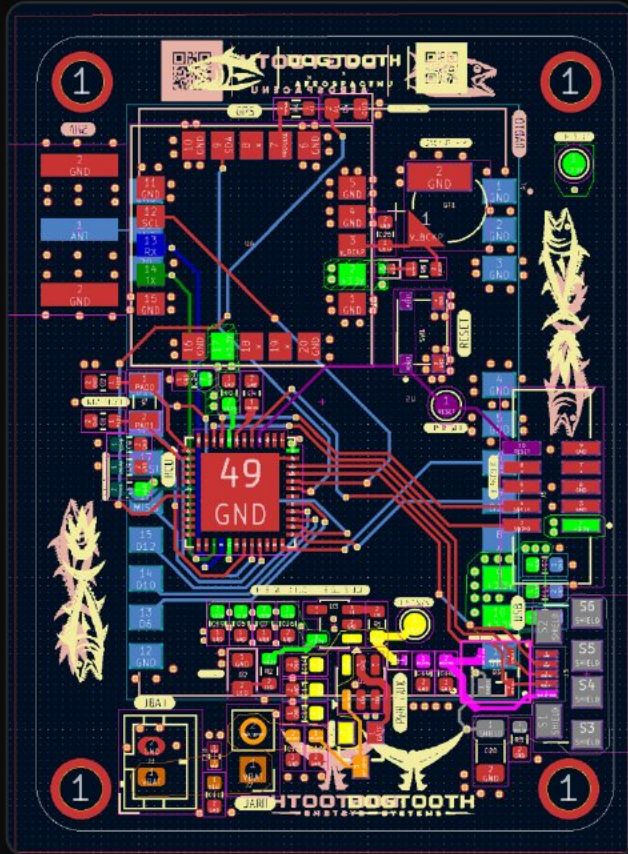
Back · zones unfilled



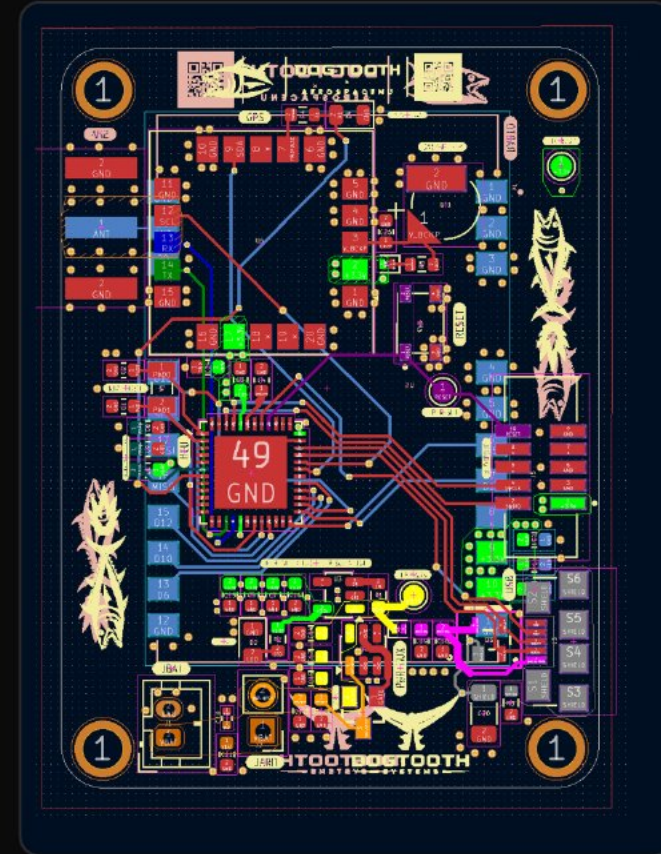
A COPPER LAYERS

Unfilled layout & RF ground

Both copper layers with zones hidden (left); the L3 GND pour outlined under the SMA (right).



Both layers · zones unfilled



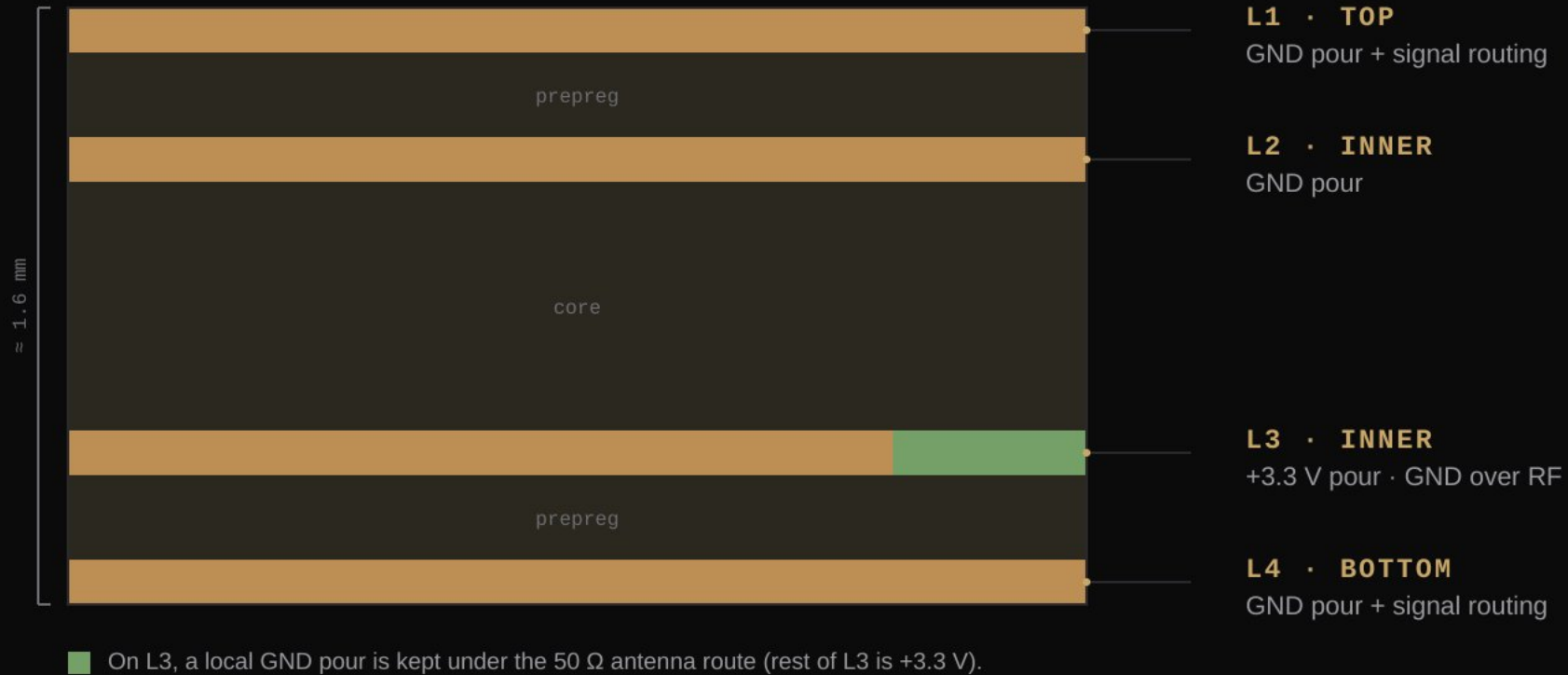
Inner · L3 GND under SMA



A FOUR-LAYER STACK

Four-layer stack-up

Copper assignment across the ≈ 1.6 mm board stack — ground-referenced throughout.



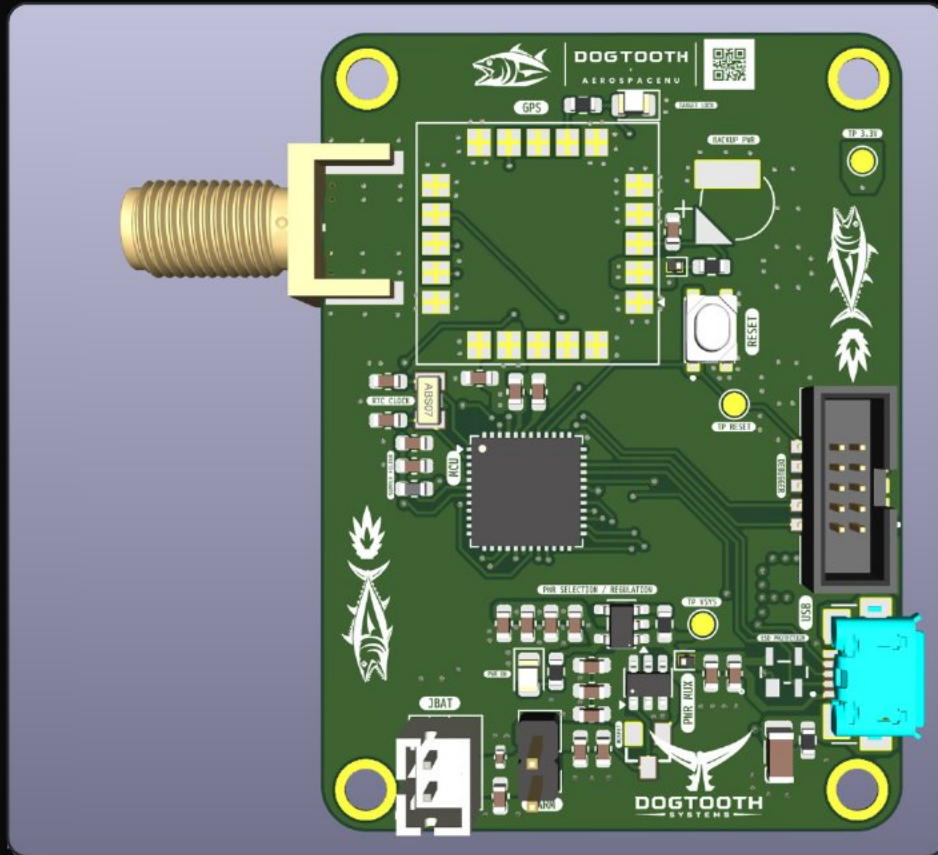


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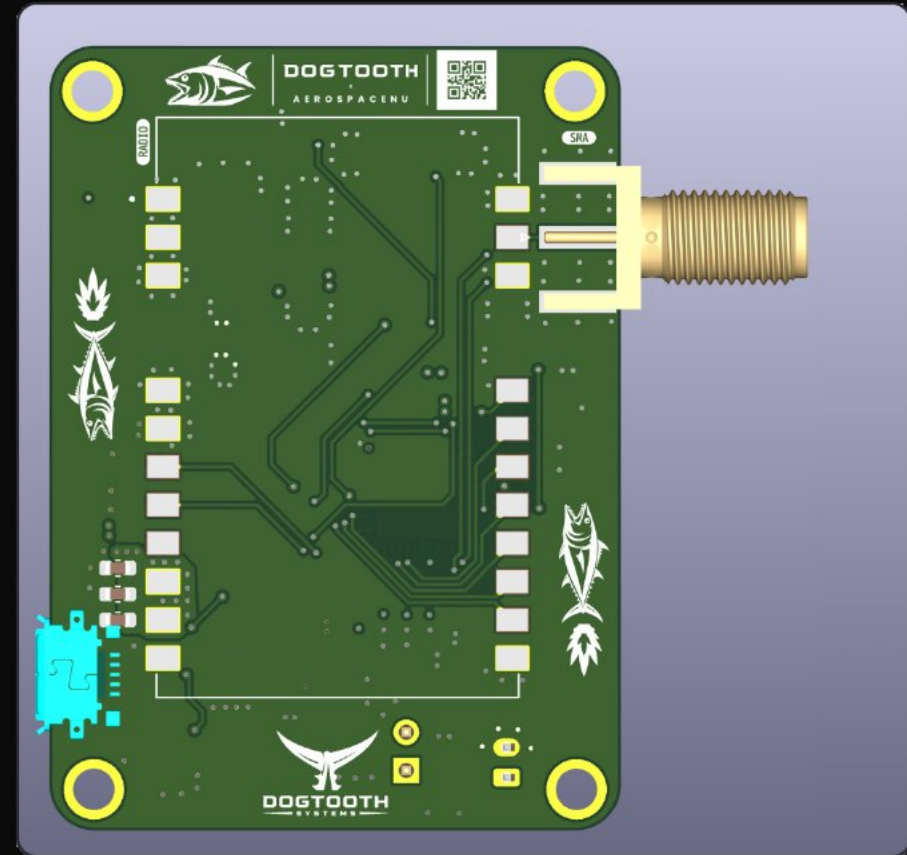
3D RENDERS

Assembled board — front & back

Rendered in the KiCad 3D viewer.



Front



Back